The Loops: An Interactive Artwork on Reclaiming Agency in Involuntary Repetition

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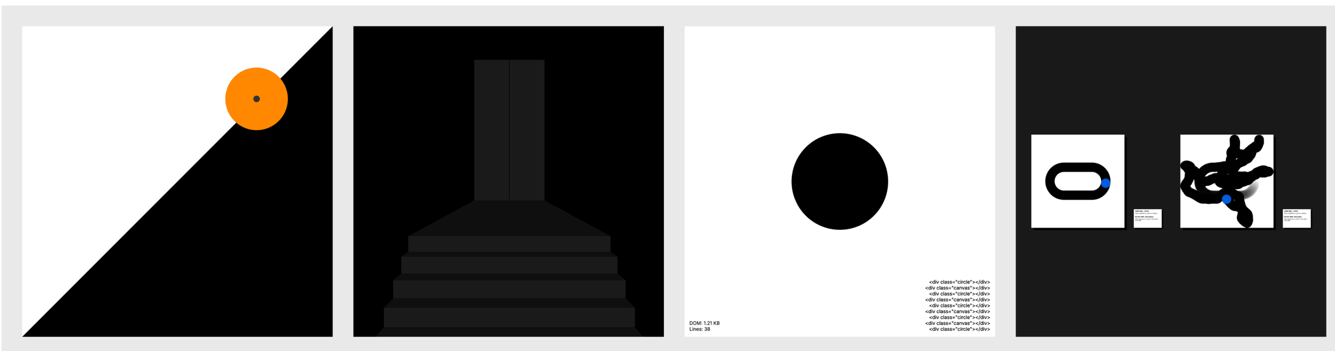

Figure 1: 4 Sketches from The Loops

Abstract

People move through many forms of involuntary repetition, from daily routines to scrolling through screens, often without noticing how these loops shape their sense of agency. When they do become aware of them, repetition can feel empty and compulsory, and it is difficult to see where choice or control might appear. This paper presents The Loops, a collection of interactive artwork on digital screens that uses simple repeated interaction to make this question tangible. Each sketch creates an abstract situation that repeats without a beginning, progress, or ending, and can be engaged through motion sensing, touch, or mouse input. The Loops explores the potential and role of interactive artwork as both a site of reflection and a medium for applied philosophy.

CCS Concepts

• **Applied computing → Media arts**; • **Human-centered computing → Activity centered design**; *Gestural input*.

Keywords

Reflective Design, Interactive Design, Media Art, Digital Art, Applied Philosophy

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1 Introduction

In contemporary life, many forms of repetition are not chosen in any strong sense [3, 4]. Work schedules, institutional routines, and personal responsibilities organize much of the day, while a second layer of digital habits forms around them: scrolling through feeds, swiping left and right, and clearing notifications in ongoing loops that are easy to enter and hard to leave [4, 10, 14]. Most of the time, these cycles stay in the background of attention. When they do become visible, they can feel empty and compulsory, and it is still unclear where a person might find a sense of choice or control inside them [13].

Studies in reflective design argue that interactive systems do more than support tasks [2, 6]. They also shape how people live and understand themselves, and can therefore be designed to invite reflection rather than only efficiency [8]. In this view, reflection means bringing usually unexamined aspects of experience into awareness so that they can become available for deliberate choice, and can be treated as a primary outcome of design [16]. Read alongside philosophical accounts of absurdity and repetition [3, 11], such as Camus's writing on Sisyphus and imposed routines, this suggests a role for interactive artifacts as a kind of applied philosophy: not to explain ideas, but to set up situations in which questions of involuntary repetition and agency can be felt and reflected on.

The Loops is an interactive artwork developed from this intersection. It consists of a collection of simple looping sketches that run on digital screens of many sizes. Each sketch constructs an abstract situation that repeats without a clear beginning, progress, or ending, and can be engaged through mouse input, touch, or motion sensing. There is no prescribed way to use the work; a loop can be a brief distraction, a small game, or a slow, contemplative ritual. The aim is to offer a place where involuntary repetition can be experienced and reflected on, and where agency may be sensed and quietly reclaimed from within the loop itself.

2 Related Work

2.1 Philosophy and Technology

Recent human-computer interaction (HCI) work has drawn on existential philosophy to think about interactive systems beyond efficiency or utility [9, 12, 15]. Nguyen et al. use Viktor Frankl's ideas to frame "meaningful HCI" around freedom, responsibility, and meaning [12], while Jiang and Ahmadpour show how VR can be designed for reflection rather than immersion by structuring moments of pre-reflection and critical reflection [8]. Taken together, they suggest that interaction can be a medium for staging questions of meaning and agency. Using the absurdist philosophy as a lens, The Loops invites users to engage with the artwork to reflect on involuntary repetition and agency.

Digital games have also used repetition and difficulty as material for philosophical or reflective experience. Getting Over It with Bennett Foddy [5], for example, confronts players with an intentionally punishing climbing task where small mistakes can erase long stretches of progress, turning iterative failure and persistence into the core of the experience rather than a side effect. This design has been discussed as an exploration of frustration, perseverance, and the player's relationship to arbitrary challenge.

2.2 Reflective Design

Designers are increasingly exploring how technology can prompt people to reflect on their experiences [1, 2, 16]. Baumer reviews how reflection has been used across interactive system design and argues that many systems implicitly aim to support reflection without clearly defining what kind of reflection is intended or how it is supposed to occur [1]. The work calls for more careful articulation of reflection as "thinking about one's own experience" and for designs that deliberately structure opportunities for such thinking, instead of treating it as a vague side effect of data presentation or feedback.

Gaver and colleagues' Drift Table offers a concrete example of this stance [6]. The project introduces an electronic coffee table that slowly pans over aerial photography in response to how weight is placed on its surface. It does not optimize any task; instead, it is designed for ludic engagement, encouraging daydreaming, curiosity, and open-ended interpretation. The Loops aligns with this reflective and ludic orientation by using simple looping interactions that invite people to linger, notice how repetition feels, and consider their own agency in relation to it, without prescribing what they should conclude.

3 Design Development

The Loops consists of four distinct sketches, each requiring users to perform repetitive actions. Through motion sensing, touch input (on touch-enabled screens), or mouse interactions, the system detects these repeated gestures and updates the visual elements accordingly. The project can be viewed at https://loops.rin.kim/

This design approach aligns with technology research aimed at reflection and moments of mental rest rather than performance efficiency, referred to as "slow technology" [7]. Rather than optimizing for task completion or measurable outcomes, The Loops focuses on the phenomenological quality of remaining within simple, repetitive situations when no external markers of progress are provided. From this perspective, a system can be meaningful simply by giving people time and space to dwell within an experience.

3.1 First Sketch: Myth of Sisyphus

https://loops.rin.kim/projects/myth-of-sisyphus

Figure 2: Myth of Sisyphus controlled through hand movement.

The first sketch is a direct response to Camus's retelling of the myth of Sisyphus [3]. An orange circle represents the boulder, and a dark triangular slope suggests the hill. Visitors can push the boulder up the slope either with a mouse or with their hand tracked through the webcam using the ml5.js handPose model. When the rock reaches the top, the hill flips direction, and the scene repeats left and right.

The rock never moves by itself. Nothing happens unless a visitor decides to act by dragging or keeping their hand in motion. There is no hidden reward, score, or ending; reaching the top only triggers the familiar fall and a mirrored hill. The piece holds a constant loop and leaves it to the visitor to decide whether to keep pushing, stop, or ignore it.

In testing, people treated this loop very differently. One participant tried hard to reach the peak, failed several times, finally succeeded, watched the rock fall, and immediately walked away. Another simply kept going, pushing the rock again and again as a kind of small challenge. A third held the rock still in the middle of the slope and said that was where they wanted it to stay. These interactions make the central question visible: the system defines the pull of gravity and the hill, but the sense of agency appears in how someone chooses to stay with, resist, or suspend the repetition.

3.2 Second Sketch: Doors

https://loops.rin.kim/projects/doors

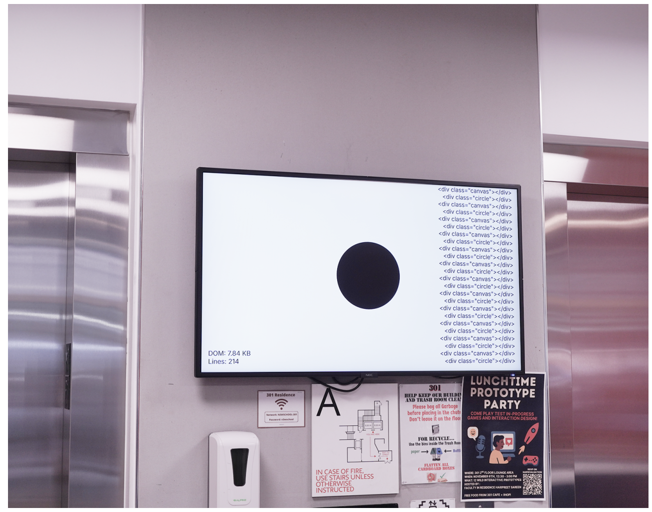
Figure 3: Doors displayed in an elevator lobby.

Doors focus on a continuous transition. The screen shows a corridor of repeating door frames and stairs that recede into depth. The environment advances forward at a steady pace, as if the viewer is walking through an endless sequence of doors. A body and face tracking model built with ml5.js estimates the upper body position of the visitor from the webcam image and uses it to rotate the corridor so that the scene subtly turns toward the person's perspective. There is no destination or final room; each door simply gives way to another corridor, and the motion continues whether or not the visitor moves.

The piece extends an everyday metaphor. A door is one of the most ordinary things we encounter, yet every doorway marks a small shift in role and identity: office to home, public to private, one phase of life to the next. In Doors, this ritual is stretched into an infinite sequence that never quite resolves. The alternating passage through frames becomes a picture of continual transformation, where every apparent ending is simply another beginning, and the viewer is kept inside the question of where they are going and who they are becoming as they keep passing through.

During testing, visitors responded to this passage in different ways. One person began to move their legs in place, trying to sync their steps with the advancing stairs. They assumed that their lower body movement was driving the scene, treating the loop as a coordinated journey. This misreading was revealing: it invited people to invent a sense of control even when none was actually available. Others sat in front of the display waiting for something more to happen, eventually expressing frustration at how slowly the corridor moved and reflecting that they've grown accustomed to fast-paced media. These reactions show how a simple, slow loop can surface expectations about pace, progress, and control, and how those expectations become material for reflection.

3.3 Third Sketch: Accumulation



https://loops.rin.kim/projects/accumulation

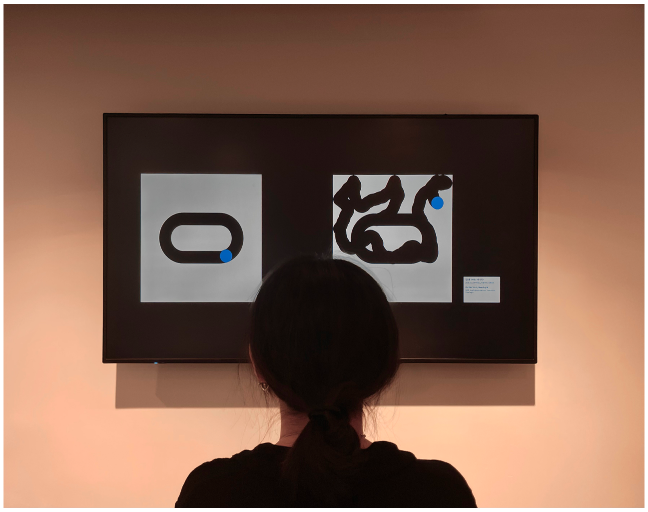
Figure 4: Accumulation displayed in an elevator lobby.

Built on JavaScript, Accumulation is based on a painting exercise in which black circles are repeatedly drawn and then covered with white paint until the canvas appears blank but grows heavier with each layer. The sketch recreates this logic in code. Every second, the sketch alternates between adding a black-circle element and adding a full-screen white-overlay. The process then repeats, sometimes placing a white circle on top of a black field. Nothing is erased; new elements are continuously stacked in the Document Object Model (DOM), so that each moment adds both a visual layer and a line of code, slowly increasing the "weight" of the page.

Visitors can click anywhere on the window to intervene. Each click resizes the next circle so that its diameter matches the distance between the click position and the center of the screen, and also flips the color relationship, producing white circles on black layers. The interaction is simple, but it lets people steer how and where the accumulation happens. The piece makes repetition visible not as a loop that returns to the same state, but as a process that quietly builds material and complexity underneath an apparently minimal surface.

In observation, this interaction was not immediately legible to most viewers. Without explanation, many people treated Accumulation as a non-interactive animation and did not discover that clicking changed the structure. The underlying idea of DOM growth as a kind of digital weight was also most apparent to visitors already familiar with HTML and the document model. However, once the background was explained, that every second adds new elements and "weight" both visually and in code, many people reported that the piece resonated with their own sense of a hidden build-up process. This gap between initial perception and later understanding became part of the piece, highlighting how some forms of repetition and accumulation remain invisible until they are named.

3.4 Fourth Sketch: Meaningful/Meaningless



https://loops.rin.kim/projects/meaningful-meaningless

Figure 5: Meaningful/Meaningless displayed on a wall-mounted screen.

This sketch grew out of a drawing exercise in which participants were asked to depict patterns of "meaningful" and "meaningless." Many drew structured, symmetrical circular forms for meaningfulness and random wandering scribbles for meaninglessness. The final piece translates these patterns into two canvases and deliberately reverses their labels. The canvas that visually resembles the "meaningful" drawings is titled **Meaningless**, while the one that resembles the "meaningless" scribbles is titled **Meaningful**, prompting viewers to question what actually feels meaningful rather than accepting the visual stereotype.

In **Meaningless**, a blue dot moves around a fixed, stadium-shaped track. The motion appears organized and controlled at first, but its endless repetition gradually suggests being stuck in a closed loop. In **Meaningful**, a blue dot roams freely across the screen. Its path is unpredictable and initially seems chaotic, yet over time, it invites curiosity about where it will go next. The white field, dark trails, and moving dot form a simple metaphor of life, path, and self, contrasting closed structure with open wandering.

Installed on a large screen in a gallery-like setting, the piece became the most debated work in the collection. Viewers discussed which side they personally agreed with and why. Many found themselves paying closer attention to Meaningful, watching the wandering dot to see where it would move next, and starting to recognize shapes or objects in the traces it left behind. This response suggests that, despite the simple visuals, the piece draws people into actively interpreting the loops, and that meaning emerges as much from this ongoing negotiation as from the motion on the screen itself.

4 Limitations

Although this exploration has opened reflections and discussions, it has several limitations. First, the work has only been experienced by a small number of people in controlled or semi-controlled settings. The interactions and quotes described here come from informal observation rather than a structured study, so they suggest possible interpretations but cannot be treated as representative of how a broader audience would respond.

Second, The Loops currently runs on a limited set of screen based setups, with reliance on webcam based tracking and browser performance. This constrains where and how the work can be installed and makes it sensitive to hardware and network stability. Some sketches, such as Accumulation, also depend on familiarity with concepts like the DOM to fully grasp the underlying idea, which narrows the audience that can access the intended meaning without additional explanation. These constraints shape both who the work reaches and what kinds of reflection it currently supports.

5 Conclusion and Future Work

The Loops presents an interactive artwork that treats involuntary repetition and agency as its primary material. Informed by reflective design and absurdist philosophy, the collection explores how simple, repeating situations can be held open without clear goals, rewards, or endings. The four sketches described here — Myth of Sisyphus, Doors, Accumulation, and Meaningful/Meaningless — offer different perspectives on repetition: effort that always resets, continuous thresholds, slow layered build-up, and the inversion of what is assumed to be meaningful or meaningless. Across these pieces, interaction is used less to accomplish tasks and more to make the structure of a loop tangible so that people can sense how it feels to inhabit it.

Informal observations suggest that even minimal loops can surface distinct stances toward repetition. Some visitors treated the work as a challenge to beat, some walked away once they felt pointless, and others chose to hold or suspend the loop in ways that made their own sense of agency explicit. In Doors and Meaningful/Meaningless, people also reflected verbally on pace, attention, and the kinds of routines that feel meaningful or empty in their own lives. While these reactions are not the result of a controlled study, they indicate that an interactive artwork can function as a site where questions of repetition, expectation, and control are articulated rather than silently endured.

Future work will focus on contextual deployments and more systematic study. One direction is to install The Loops in locations already shaped by involuntary repetition, such as office lobbies, school corridors, waiting areas, or transit spaces, and to observe how people encounter the work when it appears as part of the everyday environment rather than as an explicit art event. Another direction is to explore multi-screen or distributed configurations, where different loops appear on different displays and visitors move between them over time, turning physical navigation itself into part of the experience of thresholds and agency.

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